MAP Approximation for MLP Example

$$P(y|\underline{x}, Y, X) \sim \int \exp(-\beta e(y, \underline{x}, \underline{w})) \exp(-\beta R(\underline{w}, Y, X))$$

$$\sim \exp(-\beta e(y, \underline{x}, \underline{w}^*)) \int d\underline{w} \exp\Big\{-\beta \sum_i \frac{\partial e}{\partial w_i}(w_i - w_i^*) - \frac{\beta}{2}\sum_{ij}(w_i - w_i^*)\underbrace{\frac{\partial^2 R}{\partial w_i \partial w_j}}_{H_{ij}}(w_j - w_j^*)\Big\} \quad (I)$$

all terms which are independent of y including $\exp(-\beta R(\underline{w}^*; Y, X))$

pos. definite & symmetric

Calculation of the integral

transformation into the eigenbasis of the Hessian $\underline{H}$:

$\rightarrow \underline{S}$: orthonormal, $\underline{S}^T\underline{S} = \underline{1}$, functional determinant of transformation: 1

$$\hat{\underline{w}} = \underline{S}^T \underline{w} \qquad \& \qquad \underline{\Lambda} = \underline{S}^T \underline{H}\, \underline{S} \quad (\text{diagonal})$$

$$\underline{F} = \underline{S}^T \underline{f}$$

$$I = \int d\hat{\underline{w}} \exp\Big\{-\beta \sum_i F_i \underbrace{(\hat{w}_i - \hat{w}_i^*)}_{\tilde{w}_i} - \frac{\beta}{2}\sum_i \Lambda_{ii} \underbrace{(\hat{w}_i - \hat{w}_i^*)^2}_{\tilde{w}_i}\Big\}$$

$$I = \prod_i \int d\tilde{w}_i \exp\Big\{-\beta F_i \tilde{w}_i - \frac{\beta}{2}\Lambda_{ii}\tilde{w}_i^2\Big\}$$

completing the square in the exponent:

$$\frac{\beta}{2}\Lambda_{ii}\Big(\tilde{w}_i^2 + 2\frac{F_i}{\Lambda_{ii}}\tilde{w}_i\Big) = \frac{\beta}{2}\Lambda_{ii}\Big\{\Big(\tilde{w}_i - \frac{F_i}{\Lambda_{ii}}\Big)^2 - \frac{F_i^2}{\Lambda_{ii}^2}\Big\}$$

$$= \underbrace{-\frac{\beta}{2}\frac{F_i^2}{\Lambda_{ii}}}_{\text{in front the integral}} + \underbrace{\frac{\beta}{2}\Lambda_{ii}\Big(\tilde{w}_i + \frac{F_i}{\Lambda_{ii}}\Big)^2}_{\text{to be integrated over}}$$

contributes with a constant

$$I \sim \prod_i \exp\Big(\frac{\beta}{2}\frac{F_i^2}{\Lambda_{ii}}\Big) = \exp\Big\{\frac{\beta}{2}\sum_i F_i \Lambda_{ii}^{-1} F_i\Big\}$$

backtransformation: $\underline{S}^T\underline{S}$ $(=1)$

$$\sim \exp\Big\{\frac{\beta}{2}\underline{f}^T \underline{H}^{-1}\underline{f}\Big\}$$

predictive distribution

$$P(y|\underline{x}; Y, X) \sim \exp\Big\{\frac{\beta}{2}\underline{f}^T\underline{H}^{-1}\underline{f} - \beta e\Big)$$

$$\frac{\partial e}{\partial \underline{w}}$$